```
!pip install gensim
import gensim
import gensim.downloader as api
from gensim.models import KeyedVectors
import numpy as np
import pandas as pd
import nltk
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.0)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 48.6 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
import gensim.downloader as api

# Load Google News Word2Vec
word2vec_model = api.load("word2vec-google-news-300")

print("Vocabulary Size:", len(word2vec_model))
print("Vector Dimension:", word2vec_model["king"].shape)
print("Vector for 'king':", word2vec_model["king"][:10])
```

```
Vocabulary Size: 3000000
Vector Dimension: (300,)
Vector for 'king': [ 0.12597656  0.02978516  0.00860596  0.13964844 -0.02563477 -0.03613281
  0.11181641 -0.19824219  0.05126953  0.36328125]
```

```
!wget http://nlp.stanford.edu/data/glove.6B.zip
!unzip -q glove.6B.zip

from gensim.models import KeyedVectors

glove_model = KeyedVectors.load_word2vec_format(
    "glove.6B.50d.txt",
    binary=False,
    no_header=True
)

print("Vocabulary Size:", len(glove_model))
print("Vector Dimension:", glove_model["king"].shape)
```

```
--2026-02-10 04:17:32--  http://nlp.stanford.edu/data/glove.6B.zip
Resolving nlp.stanford.edu (nlp.stanford.edu)... 171.64.67.140
Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:80... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://nlp.stanford.edu/data/glove.6B.zip [following]
--2026-02-10 04:17:32--  https://nlp.stanford.edu/data/glove.6B.zip
Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip [following]
--2026-02-10 04:17:32--  https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip
Resolving downloads.cs.stanford.edu (downloads.cs.stanford.edu)... 171.64.64.22
Connecting to downloads.cs.stanford.edu (downloads.cs.stanford.edu)|171.64.64.22|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 862182613 (822M) [application/zip]
Saving to: ‘glove.6B.zip’

glove.6B.zip        100%[===================>] 822.24M  5.00MB/s    in 2m 41s

2026-02-10 04:20:13 (5.10 MB/s) - ‘glove.6B.zip’ saved [862182613/862182613]

Vocabulary Size: 400000
Vector Dimension: (50,)
```

```
pairs = [
    ("doctor","nurse"),
```

```
    ("cat","dog"),
    ("car","bus"),
    ("king","queen"),
    ("teacher","student"),
    ("india","china"),
    ("computer","keyboard"),
    ("apple","mango"),
    ("sun","moon"),
    ("city","village")
]

for w1, w2 in pairs:
    print(w1, "-", w2, ":", word2vec_model.similarity(w1, w2))
```

```
doctor - nurse : 0.63195235
cat - dog : 0.76094574
car - bus : 0.4693371
king - queen : 0.6510957
teacher - student : 0.63013655
india - china : 0.35330796
computer - keyboard : 0.39639163
apple - mango : 0.57518554
sun - moon : 0.4262834
city - village : 0.47896868
```

```
words = ["king", "university", "hospital", "india", "computer"]

for w in words:
    print("\nNearest words to:", w)
    print(word2vec_model.most_similar(w, topn=5))
```

```
Nearest words to: king
[('kings', 0.7138045430183411), ('queen', 0.6510956883430481), ('monarch', 0.6413194537162781), ('crown_prince', 0.6204220056533

Nearest words to: university
[('universities', 0.7003918886184692), ('faculty', 0.6780907511711121), ('unversity', 0.6758289933204651), ('undergraduate', 0.6

Nearest words to: hospital
[('Hospital', 0.7931912541389465), ('hopsital', 0.7784007787704468), ('hosptial', 0.7581779956817627), ('hospitals', 0.721253812

Nearest words to: india
[('indian', 0.6967039704322815), ('usa', 0.6836211085319519), ('pakistan', 0.681516706943512), ('chennai', 0.6675503253936768),

Nearest words to: computer
[('computers', 0.7979379892349243), ('laptop', 0.6640493273735046), ('laptop_computer', 0.6548868417739868), ('Computer', 0.6473
```

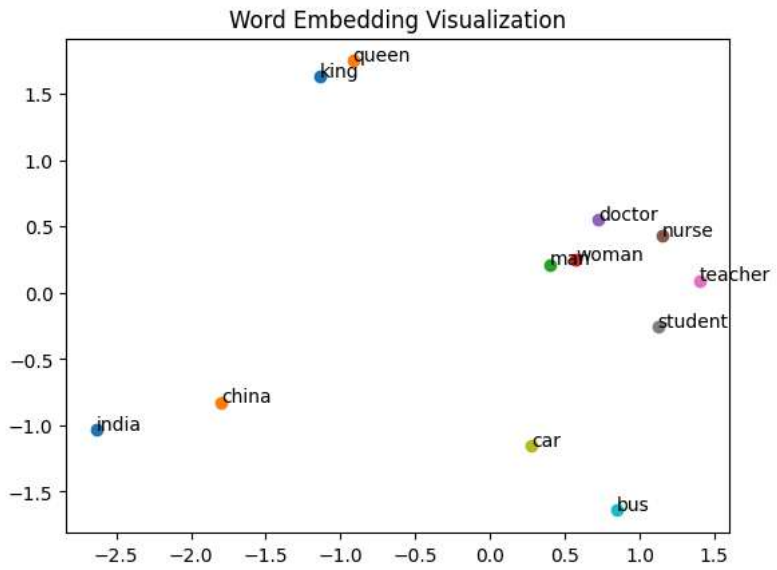
```
print("king - man + woman =")
print(word2vec_model.most_similar(
    positive=["king","woman"],
    negative=["man"],
    topn=1
))

print("paris - france + india =")
print(word2vec_model.most_similar(
    positive=["india","paris"],
    negative=["france"],
    topn=1
))
```

```
king - man + woman =
[('queen', 0.7118193507194519)]
paris - france + india =
[('chennai', 0.5442505478858948)]
```

```
words = ["king","queen","man","woman","doctor","nurse",
         "teacher","student","car","bus","india","china"]

vectors = [word2vec_model[w] for w in words]

pca = PCA(n_components=2)
reduced = pca.fit_transform(vectors)
```

```
plt.figure()
for i, word in enumerate(words):
    plt.scatter(reduced[i,0], reduced[i,1])
    plt.text(reduced[i,0], reduced[i,1], word)

plt.title("Word Embedding Visualization")
plt.show()
```